PROJECT BASED LEARNING (PBL) – PHASE 2

PROJECT PROPOSAL REPORT

PROJECT AND TEAM INFORMATION

Project Title: GEU Student Skill Exchange System

Course: B.Sc. IT **Semester:** 3rd Semester

Institution: Graphic Era Deemed to be University

### Student / Team Information • Last name: (Masoh ,Gloria, student ID# 25132008,email:taitanminawaley@gmail.com,picture) • Course: B.Sc. IT (Semester 3) ## Project Type: Individual PBL Project	

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

The GEU Student Skill Exchange System has two main objectives: to understand the techniques applied to utilize relational databases and data structures in resolving real-world skill-sharing challenges faced by university students.

Currently, students who want to learn academic or practical skills (such as programming, web development, graphic design, or public speaking) rely on informal word-of-mouth or social media groups. This makes it difficult to discover suitable learning partners efficiently.

By leveraging structured database storage and dynamic data structures (arrays and linked lists), this system provides a centralized platform for students to match, teach, and learn skills from their peers.

State of the Art / Current Solution (1 pt)

Existing peer learning methods rely on unstructured messaging groups (e.g., WhatsApp, Telegram) or personal contacts. These methods lack a central registry, making skill search time-consuming and unreliable.

The proposed system introduces a web-based portal backed by a MySQL database and in-memory data structures. It enables precise matching between skills students can teach and skills they want to learn, alongside a structured exchange-request workflow.

Project Goals and Milestones (2 pts)

1. Centralized Skill Database: Store and maintain student profiles, teachable skills, and learning requirements.
2. Skill Matching & Discovery: Implement search logic to connect students offering a skill with those seeking to learn it.
3. Request Management System: Enable students to send, accept, or reject peer skill-exchange requests.
4. Data Structures Integration: Utilize arrays and linked lists for temporary collection handling and search result processing.
5. Database Operations: Perform SQL CRUD operations across relational tables (students, skills, student_skills, exchange_requests).
6. Learning by Doing: Develop practical skills in database management, web application design, and system architecture.

7. Interactive Prototype: Deliver a working functional prototype demonstrating student skill matching.

PROJECT APPROACH (3 pts)

8. Requirement Analysis: Identifying core entities and functional requirements including student registration, skill tag classification, and request handling.
9. Database & Schema Design: Structuring relational tables using primary and foreign key constraints in MySQL.
10. Choice of Data Structures: Using arrays for dynamic search results and linked lists to manage sequential user queues and active requests.
11. System Implementation: Building the web interface and application logic to process insertion, retrieval, and request updates.
12. Testing and Validation: Verifying user creation, search accuracy, and request status updates through test cases.
13. Results and Evaluation: Assessing system correctness and efficiency in connecting peer learners.

SYSTEM ARCHITECTURE (High Level Diagram) (2 pts)

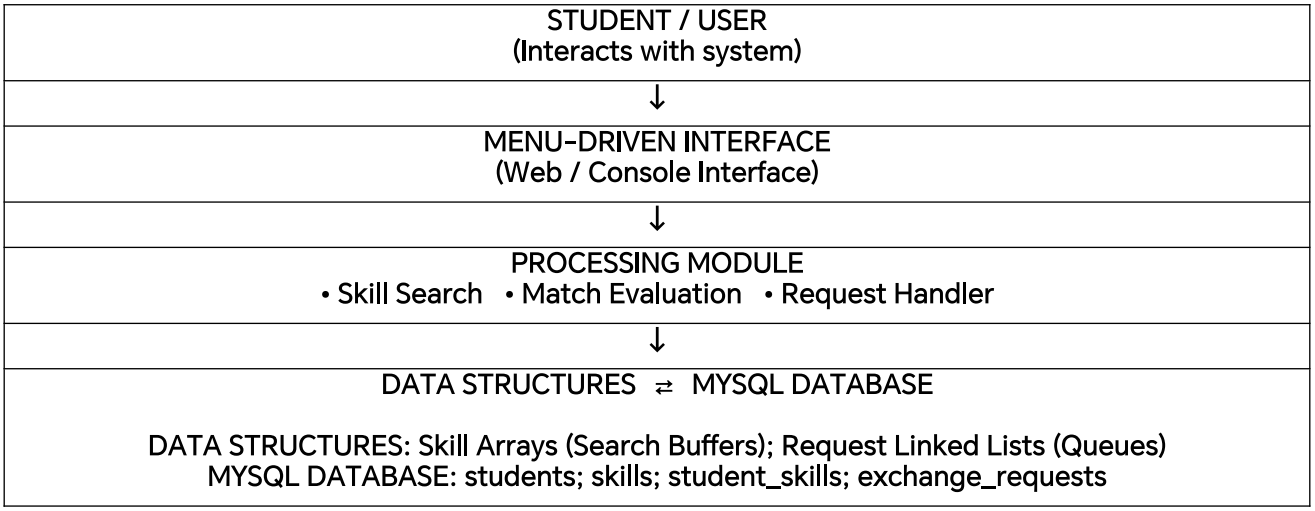


Fig 3.1: System Architecture Diagram

PROJECT OUTCOME / DELIVERABLES (1 pt)

14. Student Profile Management: Features for registering, updating, and viewing student profiles and skills.
15. Skill-Based Discovery: Direct filtering and identification of learning partners based on skills offered.
16. Request Lifecycle Management: Tracking exchange requests through 'Pending', 'Accepted', and 'Rejected' states.
17. Relational Database Integration: Schema implementation in MySQL with foreign key

constraints.

18. In-Memory Operations: Integration of array and linked list structures for data processing.
19. Technical Experience: Practical hands-on learning in DBMS, SQL queries, system design, and web architecture.

ASSUMPTIONS

- Each student has a unique Student ID and institutional email address.
- Skills are categorized into predefined domain tags (e.g., Programming, Design).
- A student can simultaneously act as both a teacher for one skill and a learner for another.
- Request status updates (Accepted/Rejected) are managed directly by the recipient student.
- The web application environment maintains persistent records within the relational database.
- The initial deployment targets internal university learning networks.

REFERENCES

20. Oracle, "MySQL 8.4 Reference Manual," MySQL Documentation.
21. PHP Documentation Group, "PHP Manual," PHP Documentation.
22. R. Elmasri and S. B. Navathe, Fundamentals of Database Systems, Pearson.
23. A. Silberschatz, P. B. Galvin, and G. Gagne, Operating System Concepts, Wiley.